**Birla Institute of Technology and Science, Pilani**
**First Semester, 2015-16**
**CHE F212,** Fluid Mechanics

Quiz 1(closed Book)
15 min, 5 points
NAME: ID:

12/08/15
Section:

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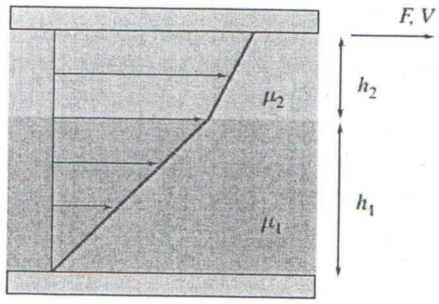
Fluids of viscosities $\mu_1 = 0.1$ N.s/m$^2$ and $\mu_2 = 0.15$ N.s/m$^2$ are contained between two plates (each plate is 1 m$^2$ in area) The thicknesses are $h_1 = 0.5$ mm and $h_2 = 0.3$ mm, respectively. Find the force F to make the upper plate move at a speed of 1 m/s. What is the fluid velocity at the interface between the two fluids?




Sol$^n$ The shear stress is same throughout (the velocity gradients are linear, and the stresses in the fluid at the interface must be equal and opposite).

Hence.

$$\tau = \mu_1 \frac{du_1}{dy} = \mu_2 \frac{du_2}{dy} \rightarrow 1$$

or

$$\mu_1 \frac{V_i}{h_1} = \mu_2 \left(\frac{V - V_i}{h_2}\right) \rightarrow 1$$

where $V_i$ is the interface velocity.

Solving for the interface velocity $V_i$

$$V_i = \frac{V}{1 + \frac{\mu_1}{\mu_2} \cdot \frac{h_2}{h_1}} = \frac{1 \text{ m/s}}{1 + \frac{0.1}{0.15} \cdot \frac{0.3}{0.5}} \rightarrow 1$$

$$V_i = 0.714 \frac{m}{s}$$

Then the force required is : $F = \tau \cdot A$

$$: \mu_1 \frac{V_i}{h_1} \cdot A = 0.1 \frac{N \cdot s}{m^2} \times 0.714 \frac{m}{s} \times \frac{1}{0.5 mm} \times \frac{1000 mm}{1 m} \times 1 m^2$$

$$\boxed{F = 143 N} \rightarrow 2$$